Effects of N-3 on metabolic syndrome: Supplementary data

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1 Introduction

This document includes supplementary data and computational details related to the paper "Effects of …".

1.1 Project organisation

1.1.1 Chapters

1.1.2 R scripts

1.1.3 Stan models

2 Analytic approach and data description

2.1 Data analysis

Descriptive data are presented as mean and standard deviations (SD).

Bayesian models were constructed using the Stan probabilistic programming language (Carpenter et al. 2017) and fitted using the `cmdstanr` interface (Gabry et al. 2025) in R (R Core Team 2025). All sampling from the prior and joint posterior distribution was done using the Hamiltonian Monte Carlo (HMC) no-U-turn sampler (Stan Reference Manual: MCMC sampling). A general workflow was adopted where models were incrementally built and tested for validity through simulations. First, consequences of prior distributions were evaluated using prior predictive simulations. Priors where selected to constrain model predictions and parameter values to *a priori* plausible ranges. Samples from the joint posterior distribution where used to create posterior predictive checks where model implications where simulated and compared to observed data (Gelman et al. 2014, 141). Performance of the HMC algorithm was assessed by inspecting chain convergence statistics (R-hat) and visually inspecting convergence and stationary sampling across multiple HMC chains.

2.2 Description of the data set

The overall objective of the present analysis is to determine the effect of N-3 supplementation in combination with resistance training on indications of metabolic health. Main outcome variables

2.2.1 Participants

Table 2.1: Participant characteristics.

	N-3 (n = 32; n = 42)	Placebo (n = 35; n = 38)
Male		
Age (years)	47 (8) [29 - 60]	49 (6) [36 - 59]
Stature (cm)	180 (8) [153 - 193]	181 (6) [165 - 192]
Body mass (kg)	96 (17) [69 - 152]	98 (18) [71 - 142]
Body mass index (kg×m^-2^)	29 (5) [20 - 45]	30 (6) [23 - 52]
Female		
Age (years)	45 (8) [30 - 59]	47 (9) [23 - 60]
Stature (cm)	169 (6) [158 - 182]	168 (6) [152 - 180]
Body mass (kg)	83 (17) [58 - 136]	79 (14) [53 - 114]
Body mass index (kg×m^-2^)	29 (6) [18 - 45]	28 (5) [20 - 40]

Data are presnted as Mean (SD) [Min-Max]

3 Effects of N-3 supplementation of serum fatty acids

Effects of N-3 supplementation on indications of metabolic syndrome

Carpenter, Bob, Andrew Gelman, Matthew D. Hoffman, Daniel Lee, Ben Goodrich, Michael Betancourt, Marcus Brubaker, Jiqiang Guo, Peter Li, and Allen Riddell. 2017. "Stan: A Probabilistic Programming Language." *Journal of Statistical Software* 76 (1): 1–32. https://doi.org/10.18637/jss.v076.i01.

Gabry, Jonah, Rok Cesnovar, Andrew Johnson, and Steve Bronder. 2025. *Cmdstanr: R Interface to 'CmdStan'*. https://mc-stan.org/cmdstanr/.

Gelman, Andrew, John B. Carlin, Hal Steven Stern, David B. Dunson, Aki Vehtari, and Donald B. Rubin. 2014. *Bayesian Data Analysis*. Third edition. Boca Raton: CRC Press.

R Core Team. 2025. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. https://www.R-project.org/.